Math 241, F18 Name: Answer

Quiz 9

<u>Instructions:</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.**

Problem 1. (5 points) Use **Lagrange multipliers method** to find the maximum value of $f(x,y) = 49 - x^2 - y^2$ on the line $x + 3y = 10$.

$$\max f(x,y) = 49 - x^2 - y^2$$

$$\text{s.t.} \quad g(x,y) = x + 3y = 10$$

We need solve

$$\begin{cases} \nabla f(x,y) = \lambda \nabla g(x,y) \\ g(x,y) = 10 \end{cases}$$

$$\Rightarrow \begin{cases} -2x = \lambda(1) \\ -2y = \lambda(3) \\ x + 3y = 10 \end{cases} \Rightarrow \begin{cases} x = \frac{\lambda}{(-2)} \\ y = \frac{3\lambda}{(-2)} \end{cases}$$

$$-\frac{\lambda}{2} + 3 \cdot \frac{3\lambda}{(-2)} = 10$$

$$-5\lambda = 10 \Rightarrow \lambda = -2$$

Thus $x = -\frac{\lambda}{2} = 1$

$y = -\frac{3}{2}\lambda = 3$

the maximum value is

$$f(1,3) = 49 - 1^2 - 3^2 = 39.$$

Problem 2. (5 points) Calculate the iterated integral $\int_0^1 \int_1^2 (4x^3 - 9x^2y^2)\, dy dx$

$$\int_0^1 \int_1^2 (4x^3 - 9x^2y^2)\, dy dx$$

$$= \int_0^1 \left[4x^3y - 3x^2y^3\right]_{y=1}^{y=2} dx = \int_0^1 [8x^3 - 24x^2] - [4x^3 - 3x^2] dx$$

$$= \int_0^1 4x^3 - 21x^2 dx = \left[x^4 - 7x^3\right]_{x=0}^{x=1}$$

$$= -6$$